



UNITED INTERNATIONAL UNIVERSITY (UIU)

Department of Computer Science and Engineering (CSE)

FINAL EXAMINATION
DURATION: 2 HOURS

FALL, 2025
FULL MARKS: 40

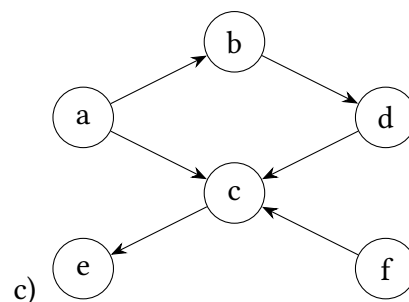
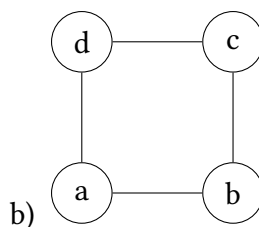
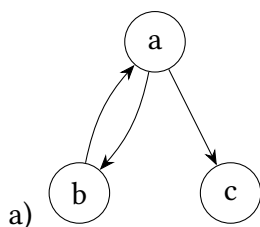
CSE 2215: Data Structures and Algorithms I

Answer **all 7 (seven)** questions. Figures in the right margin indicate full marks of questions.

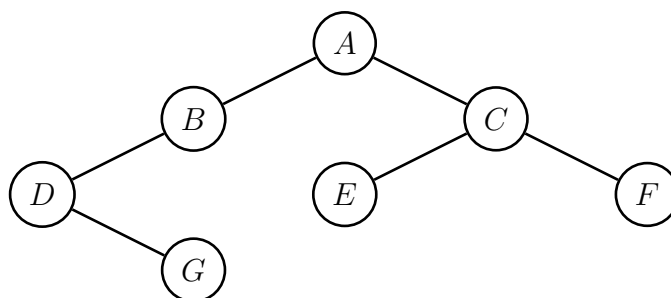
[Any examinee found adopting unfair means will be expelled from the trimester/program according to UIU disciplinary rules.]

You do not need to write any programming code for any question in this examination.

1. For each of the following graphs, find out **how many zeroes** are in its **adjacency matrix**. 3
You **do not need** to draw the graphs or show the steps.



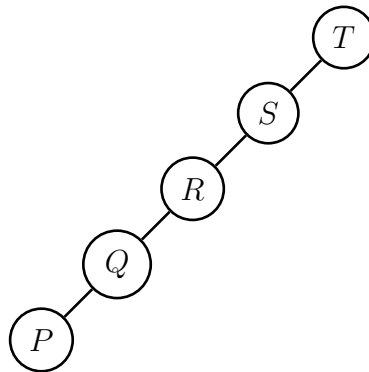
2. a) Draw the **complete binary tree** with the following array representation: [10, 20, 30, 40, 50, 60]. 1
b) Show the simulation of the **traversal** of the following binary tree starting at the root(A), visiting closest nodes first (left to right) before moving deeper. 3



- c) Represent the values of the tree using an **array**. Some elements of the array may be NULL. 3
You **do not need** to draw the tree or show the steps.
3. a) Why does **topological sort** only work on a Directed Acyclic Graph (DAG)? 2
b) Draw the **diagram** of a DAG to represent the prerequisite relationships among the courses described in the following table. In the graph, course codes should be used as **vertices** and there should be a **directed edge** from each prerequisite course to the dependent course. For example, one of the edges should be $CSE\ 1110 \rightarrow CSE\ 1111$. 2

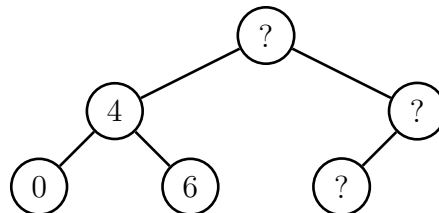
Course Code	Prerequisite(s)
CSE 1110	-
CSE 1111	CSE 1110
CSE 2215	CSE 1111
CSE 2217	CSE 2215
CSE 3811	CSE 2217, MATH 2205
MATH 2205	-

- c) Find a valid topological order for the graph or show that it doesn't exist. 3
4. Answer the following questions **sequentially** (use the answer of one question to answer the next). 2
- a) Build a **min-heap** from the following array: [10, 50, 40, 70, 60, 80]. 2
- b) Insert the value 20 in the heap and show the result. 2
- c) Simulate the first 2 steps of **heap sort** (deletion of the 2 smallest elements) on the current heap. 3
- d) Choose the faster algorithm between **heap sort** and **selection sort** in terms of time complexity. Explain your answer in 1-3 sentences. 2
5. a) Among **pre-order**, **in-order**, and **post-order** traversal sequences, which will sort the nodes of the following tree in **lexicographical order** (ascending) of their values? 2



If there are multiple correct answers, mention **all** of them.
You **do not need** to draw the tree or show the steps.

- b) Construct the **tree** that has the following traversal sequences (only show the final tree). 2
In-order: [1, 17, 25, 32]
Pre-order: [1, 25, 17, 32].
- c) Check out the following tree. Here the nodes marked with '?' have unknown values. 2



Is it possible that the **pre-order** traversal sequence of this tree be [2, 4, 3, 0, 5, 6]? Explain your answer briefly.
You **do not need** to draw the tree.

6. a) What is a BST (Binary Search Tree)? 1
- b) Construct a BST with the following elements: ['map', 'ink', 'sun', 'cup', 'pen', 'sky', 'box', 'net'] and show the result. 2

- c) **Insert** the value 'jet' into the BST and **delete** 'cup'.
Only show the final updated tree.

3

7. Which traversal will be more suitable between **BFS (Breadth First Search)** and **DFS (Depth First Search)** to reach from Start node to End node on the following graph? Explain Briefly.

2

